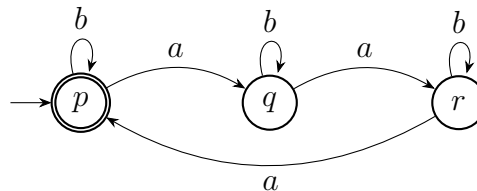


## Quiz 1

CPSC 406 — Algorithm Analysis,  
Chapman University · Spring 2026  
Instructor: Jonathan Weinberger

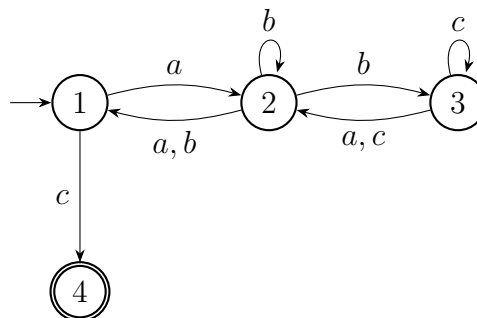
Name: \_\_\_\_\_

1. (2 points) (a) Consider the following DFA  $\mathcal{A}$  over the alphabet  $\Sigma = \{a, b\}$ :



Which of the following languages is  $L(\mathcal{A})$ ? Tick exactly one.

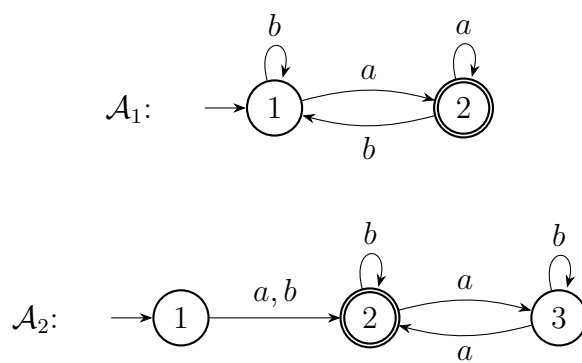
- ☐  $L_1 = \{ w \in \Sigma^* \mid |w| = 3 \}$
  - ☐  $L_2 = \{ w \in \Sigma^* \mid |w|_a = 3 \}$
  - ☐  $L_3 = \{ w \in \Sigma^* \mid |w| \equiv 0 \pmod{3} \}$
  - ☐  $L_4 = \{ w \in \Sigma^* \mid |w|_a \equiv 0 \pmod{3} \}$
  - ☐ None of these.
- (b) Consider the following NFA  $\mathcal{B}$  over the alphabet  $\Sigma = \{a, b, c\}$  with initial state 1 and accepting state 4:



Which of the following words are accepted by  $\mathcal{B}$ ? Tick exactly one.

- ☐  $aab$
- ☐  $abc$
- ☐  $ccc$
- ☐ All of these.
- ☐ None of these.

2. (2 points) Consider the following two DFAs over the alphabet  $\Sigma = \{a, b\}$ :



Draw a DFA recognizing  $L(\mathcal{A}_1) \cap L(\mathcal{A}_2)$ .